

Google HACKELTON- 2026
ATRIA INSTITUTE OF TECHNOLOGY
(Hebbal)
“EldersSeva”

Guide:

Apoorva M

Submitted By:

SHRI PHANEENDRA SHARMA PR(Team Leader)

NAWAZ AHAMMED

Varshitha S H

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Abstract :-

India witnesses large-scale internal migration driven by employment, education, and socio-economic factors. However, migration is difficult to measure in near real-time using traditional census or survey-based approaches. Aadhaar, being a ubiquitous digital identity system, records large-scale enrolment and update activities across the country, which can serve as a powerful proxy indicator for population movement and service demand.

This project proposes a data-driven analytical framework that uses anonymized Aadhaar enrolment and demographic update datasets to construct a “Migration Pressure Score (MPS)”, which helps identify districts and regions experiencing unusually high inflow or mobility-related pressure. The core idea is that adult address updates indicate relocation or mobility, while child enrolments represent relatively stable natural population growth. By comparing these two signals, we derive a relative indicator of migration-driven service demand.

Using large-scale UIDAI datasets, we perform data cleaning, aggregation, exploratory analysis, regional comparison, and time-series analysis to identify high-pressure districts and emerging trends. The results are visualized through charts and dashboards, and the insights are translated into actionable policy recommendations for public service planning, ration distribution, urban infrastructure readiness, and administrative capacity planning.

This work demonstrates how anonymized administrative data can be transformed into a decision-support system for proactive governance while fully respecting privacy and data protection principles.

1.Introduction and Background :-

India has one of the largest and most dynamic internal migration systems in the world. Millions of people move across districts and states every year for employment, education, marriage, and other socio-economic reasons. This migration creates significant pressure on urban infrastructure, public distribution systems, healthcare, housing, and administrative services.

However, migration data is traditionally captured through census and surveys, which are infrequent, delayed, and not suitable for real-time or near-real-time policy response. As a result, administrators often react to service stress only after problems become visible on the ground.

Aadhaar, as a digital public infrastructure, records large-scale enrolment and update activities across India. While the data is anonymized and aggregated, it still reflects patterns of population interaction with the identity system. In particular, address updates by adults often correlate with relocation or mobility, and new child enrolments reflect baseline population growth.

This project explores whether Aadhaar service usage patterns can be used as a proxy signal to identify regions experiencing high migration-related pressure and thus support proactive governance and better resource planning.

2. Problem Statement :-

Government systems currently lack a near real-time, data-driven indicator to identify districts and regions experiencing high migration-driven service demand. Traditional data sources are delayed and not suitable for rapid response.

The problem addressed in this project is:

“How can anonymized Aadhaar enrolment and update data be used to construct a reliable, privacy-preserving indicator that highlights districts and regions experiencing unusually high migration or mobility pressure, so that public services and administrative capacity can be planned proactively?”

3. Objectives :-

The main objectives of this project are:

1. To analyze Aadhaar enrolment and demographic update datasets at scale.
2. To construct a Migration Pressure Score (MPS) using aggregated service usage patterns.
3. To identify districts and states with unusually high relative update activity.
4. To study temporal trends and regional clustering of migration pressure.
5. To visualize the results in a clear and interpretable manner.
6. To propose policy-relevant use cases and administrative actions based on the findings.

4. Datasets Used :-

The project uses the following anonymized and aggregated datasets provided by UIDAI:

1. **Aadhaar Enrolment Dataset** This dataset contains daily or periodic counts of Aadhaar enrolments, disaggregated by location (state, district, pincode) and age groups. In this project, the age group 0–5 is used as a proxy for new child enrolments, representing baseline population growth.
2. **Aadhaar Demographic Update Dataset** This dataset contains aggregated counts of demographic updates (such as address updates), again broken down by date and geography. In this project, the 17+ age group is used as a proxy for adult address updates, which often correlate with relocation or migration.

All datasets are anonymized, aggregated, and used strictly in compliance with UIDAI data usage guidelines. No individual-level data is used or inferred.

5. Data Understanding and Exploration :-

In the initial stage, the datasets were explored to understand:

- The total number of records
- The time range covered
- The geographical coverage
- The distribution of values across states and districts
- The presence of missing or zero values

Basic descriptive statistics and exploratory charts were generated to understand the scale and structure of the data. This step also helped confirm that the datasets are aggregated and suitable for region-level and time-series analysis rather than individual tracking.

6. Data Cleaning and Preprocessing :-

The following preprocessing steps were applied:

1. All CSV files belonging to the same dataset type were merged into single master datasets (one for enrolment and one for demographic updates).
2. Only relevant columns (date, state, district, pincode, and required age-group counts) were retained.
3. Data types were standardized (dates, numeric fields).
4. Missing values were handled by treating them as zero where appropriate.
5. Data was aggregated at district and date levels where required for analysis.

These steps ensured that the data was consistent, clean, and suitable for further analysis.

7. Methodology :-

The core idea of the methodology is to construct a relative indicator of migration-related pressure using two signals:

- Child enrolments (0–5 age group) as a proxy for baseline population growth
- Adult demographic updates (17+ age group) as a proxy for mobility or relocation activity

Based on this, we define:

Migration Pressure Score (MPS) = $\frac{\text{Adult_Demographic_Updates}_{17+}}{(\text{Child_Enrolments}_{0_5} + 1)}$

The +1 in the denominator is used to avoid division by zero.

This score is computed:

- For each district
- Over time (daily or monthly aggregates)

Higher values of MPS indicate regions where adult update activity is disproportionately high compared to baseline population growth, suggesting higher migration or mobility pressure.

8. Analysis and Results :-

The following analyses are performed:

1. Ranking districts by Migration Pressure Score to identify top high-pressure regions.
2. State-wise aggregation to identify which states are experiencing higher overall pressure.
3. Time-series analysis to see whether migration pressure is increasing or decreasing in certain regions.
4. Comparative analysis between stable regions and high-pressure regions.

The results are presented using:

- Bar charts
- Line charts
- Tables
- (Optional) Map-based visualizations

Each visualization is accompanied by an explanation describing what it shows and why it is important.

(Insert main charts and results here)

9. Validation and Limitations :-

It is important to note that:

- This system does not track individuals.
- This system does not claim to measure migration directly.
- The MPS is a proxy indicator based on service usage patterns.

Limitations include:

- Aadhaar updates may occur for reasons other than migration.
- Not all migration events necessarily result in immediate address updates.
- The data is aggregated and cannot capture micro-level individual behavior.

Despite these limitations, the indicator is still useful as a relative, early-warning, decision-support signal.

10. Impact and Applicability :-

This framework can be used by administrators to:

- Identify districts that may need additional ration supply, healthcare capacity, or infrastructure.
- Plan mobile service units or temporary centres in high-pressure regions.
- Prioritize inspections, audits, or field surveys in flagged districts.
- Support evidence-based planning instead of reactive decision-making.

11. System / Dashboard Architecture (If Applicable) :-

The project can be implemented as a simple data analysis pipeline and dashboard:

- Data Layer: CSV datasets
- Processing Layer: Python (Pandas, NumPy)

- Visualization Layer: Matplotlib / Seaborn / Streamlit
- Output: Charts, tables, and interactive dashboard

(Insert architecture diagram here)

12. User Interface and Features (If Applicable) :-

The dashboard can provide:

- Top high-pressure districts list
- State-wise comparison
- Time trend charts
- Data filters by state or date

(Insert UI screenshots here)

Appendix A: Code Implementation

(Insert full code here: data loading, cleaning, analysis, visualization, dashboard)

```
EldersSeva > > START_APP.bat
1  @echo off
2  echo =====
3  echo   STARTING ELDERS SEVA DASHBOARD (1.8M Records)
4  echo =====
5  echo.
6  echo 1. Starting Backend Server (Port 8888)...
7  start "Elders Backend" cmd /k "npm run start"
8  echo   - Waiting for backend to initialize...
9  timeout /t 5 >nul
10
11  echo.
12  echo 2. Starting Frontend Dashboard (Port 5173)...
13  start "Elders Frontend" cmd /k "npm run dev"
14  echo   - Waiting for frontend to be ready...
15  timeout /t 5 >nul
16
17  echo.
18  echo 3. Opening Dashboard in Browser...
19  start http://localhost:5173
20
21  echo.
22  echo =====
23  echo   SUCCESS! The system is running.
24  echo   Do not close the two black windows that opened.
25  echo =====
26  pause
27
```

```
EldersSeva > verification_log.txt
1  =fÜÇ Starting Verification Test...
2
3  =föæ Logging in as Admin...
4  Γfà Login successful. Token received.
5
6  ΓA| Triggering Data Load (First Call - Expect delay)...
7  ΓÄñÄ First Call Duration: 3.88s
8  Γfà Data loaded. Records in response: 0 (Should be 5000)
9
10 ΓÜi Testing Cache (Second Call - Expect instant)...
11 ΓÄñÄ Second Call Duration: 0.01s
12 Γfà CACHE IS WORKING! (Response was instant)
13
14 =föÑ Testing Full Download Stream...
15
16 Γfà Download Complete!
17   - Total Bytes: 0.00 MB
18   - Time Taken: 0.03s
19   - Speed: 0.00 MB/s
20 ΓÜäñÄ File size seems too small. Check if full data is being sent.
21
```

```
1  < import React, { useState, useEffect } from 'react';
2  < import {
3    BarChart, Bar, LineChart, Line, PieChart, Pie, ScatterChart, Scatter,
4    XAxis, YAxis, CartesianGrid, Tooltip, ResponsiveContainer, Legend, Cell
5  } from 'recharts';
6
7  const COLORS = [ '#808080', '#808080', '#808080', '#808080', '#808080', '#808080', '#808080', '#808080', '#808080', '#808080' ];
8  import { loadResult, saveResult, clearResult } from './utils/indexedDB';
9
10 < const DataAnalysis = () => {
11   const [analysisResult, setAnalysisResult] = useState(null);
12   const [currentGraph, setCurrentGraph] = useState('overview');
13   const [showDuplicates, setShowDuplicates] = useState(false);
14   const [loading, setLoading] = useState(false);
15   const [isAuthenticated, setIsAuthenticated] = useState(localStorage.getItem('authToken'));
16   const [authEmail, setAuthEmail] = useState('admin@eldersseva.com');
17   const [authPassword, setAuthPassword] = useState('');
18   const [currentPage, setCurrentPage] = useState(1);
19   const [uploadedFile, setUploadedFile] = useState(null);
20   const MAX_FILE_SIZE = 50;
21
22   // Authentication handler
23   const handleAuth = async (e) => {
24     e.preventDefault();
25     try {
26       const response = await fetch('/api/auth/login', {
27         method: 'POST',
28         headers: { 'Content-Type': 'application/json' },
29         body: JSON.stringify({ email: authEmail, password: authPassword })
30       });
31       const data = await response.json();
32       if (response.ok && data.token) {
33         localStorage.setItem('authToken', data.token);
34         localStorage.setItem('username', 'Admin');
35         setIsAuthenticated(true);
36         setAuthPassword('');
37         alert('Admin login successful!');
38         // Load analysis data
39         loadAnalysisData(data.token);
40       } else {
41         alert(data.error || 'Authentication failed');
42       }
43     } catch (error) {
44       alert('Authentication error: ' + error.message);
45     }
46   };
47
48   const handleLogout = async () => {
49     try {
50       const token = localStorage.getItem('authToken');
51       await fetch('/api/auth/logout', {
52         method: 'POST',
53         headers: {
54           'Content-Type': 'application/json',
55           'Authorization': `bearer ${token}`
56         }
57       });
58     } catch (error) {
59       console.error('Logout error:', error);
60     }
61     localStorage.removeItem('authToken');
62     localStorage.removeItem('username');
63     setIsAuthenticated(false);
64     setAnalysisResult(null);
65   };
66
67   // Smart CSV Parser & Normalizer
68   const processUploadedFile = (context) => {
69     const lines = context.split('\n');
70     if (lines.length < 2) {
71       alert('File appears to be empty or invalid.');
72       return;
73     }
74
75     const headerLine = lines[0].toLowerCase();
76     const headers = headerLine.split(',').map(h => h.trim());
77
78     // Smart Mapping Strategy
79     const map = {
```

```

83     adhaar: headers.findIndex(h => h.includes('adhaar')) || h.includes('id') || h.includes('id'),
84     name: headers.findIndex(h => h.includes('name')) || h.includes('beneficiary'),
85     state: headers.findIndex(h => h.includes('state')) || h.includes('province') || h.includes('region'),
86     district: headers.findIndex(h => h.includes('district')) || h.includes('city') || h.includes('location'),
87     pincode: headers.findIndex(h => h.includes('pin')) || h.includes('zip') || h.includes('code'),
88     age: headers.findIndex(h => h.includes('age')) || h.includes('years') || h.includes('dob'),
89     biogas: headers.findIndex(h => h.includes('bio')) || h.includes('biological'),
90     status: headers.findIndex(h => h.includes('status')) || h.includes('result') || h.includes('outcome'))
91   };
92
93   console.log('🔗 Smart Header Mapping', map);
94
95   const parsedRecords = [];
96   let successCount = 0;
97
98   for (let i = 1; i < lines.length; i++) {
99     const line = lines[i];
100     if (!line.trim()) continue;
101
102     // Handle single CSV (doesn't support quoted commas perfectly but good enough for general use)
103     const cols = line.split(',');
104
105     const record = {
106       adhaar: map.adhaar > -1 ? cols[map.adhaar] : `ROW-${i}`,
107       name: map.name > -1 ? cols[map.name] : 'Unknown',
108       state: map.state > -1 ? cols[map.state] : 'Unknown State',
109       district: map.district > -1 ? cols[map.district] : 'Unknown District',
110       pincode: map.pincode > -1 ? cols[map.pincode] : 'Unknown',
111       age: map.age > -1 ? parseInt(cols[map.age]) || 0 : 0,
112       biogas: map.biogas > -1 ? parseInt(cols[map.biogas]) || 0 : 0,
113       status: map.status > -1 ? cols[map.status] : 'Success', // Default to success if no status col
114       biometricType: 'Uploaded'
115     };
116
117     // Basic validation
118     if (record.state) record.state = record.state.trim().replace(/"/g, '');
119     if (record.district) record.district = record.district.trim().replace(/"/g, '');
120     if (record.status) record.status = record.status.trim().replace(/"/g, '');
121
122     parsedRecords.push(record);
123     successCount++;
124   }
125
126   console.log('✅ Parsed', successCount, 'records from upload');
127
128   // Integrate into existing analysis flow
129   const deduped = runDeDuplication(parsedRecords);
130   setAnalysisResult(deduped);
131   alert('Successfully loaded', successCount, 'records from your file!');
132 }
133
134 const handleFileUpload = (event) => {
135   const file = event.target.files[0];
136   if (!file) return;
137
138   console.log('📁 Processing upload:', file.name);
139   setUploadStatus('uploading');
140   setLoading(true);
141
142   const reader = new FileReader();
143   reader.onload = (e) => {
144     const text = e.target.result;
145     processFileUpload(text);
146     setLoading(false);
147   };
148   reader.onerror = () => {
149     alert('Error reading file');
150     setLoading(false);
151   };
152   reader.readAsText(file);
153 };
154
155 // Refactored DeDuplication logic to be reusable for both backend data and uploaded data
156 const runDeDuplication = (records) => {
157   // ⚡ FAST DEPLICATION USING O(1) Complexity
158   const adhaarSet = new Set();
159   const profileMap = new Map();
160   const duplicates = [];

```

```

EldersSeva > components @ StatefulService
1 import React, { useState, useEffect } from 'react';
2
3 const INITIAL MOCK DATA = [
4   {
5     id: string;
6     name: string;
7     service: string;
8     region: string;
9     status: string;
10    date: string;
11  },
12 ];
13
14 const [adhaarId, setAdhaarId] = useState('');
15 const [loading, setLoading] = useState(false);
16 const [showStatus, setShowStatus] = useState(false);
17 const [systemMessage, setSystemMessage] = useState('');
18
19 // State-driven registry with localstorage persistence
20 const [publicRegistry, setPublicRegistry] = useState([]);
21
22 // Initialize from localstorage
23 useEffect(() => {
24   const savedData = localStorage.getItem('elders_seva_registry');
25   if (savedData) {
26     try {
27       setPublicRegistry(JSON.parse(savedData));
28     } catch (e) {
29       setPublicRegistry(INITIAL MOCK DATA);
30     }
31   } else {
32     setPublicRegistry(INITIAL MOCK DATA);
33     localStorage.setItem('elders_seva_registry', JSON.stringify(INITIAL MOCK DATA));
34   }, []);
35
36 const mockSteps = [
37   { title: 'Application Submitted', description: 'Exception request registered in the National Portal.', date: 'Oct 12, 2025', isCompleted: true, isCurrent: false },
38   { title: 'Technical Audit', description: 'Automated check completed. Biometric mismatch confirmed due to age-related friction loss.', date: 'Oct 14, 2025', isCompleted: true, isCurrent: false },
39   { title: 'Manual Field Verification', description: 'Regional officer is currently verifying physical foundational documents.', date: 'Oct 16, 2025', isCompleted: false, isCurrent: true },
40   { title: 'Final Disbursement', description: 'Authorized override for service activation.', date: '-', isCompleted: false, isCurrent: false },
41 ];
42
43 // Enhanced resource allocation data
44 const resourceAllocation = [
45   { service: 'Old Age Pension', allocated: true, amount: '₹1,800/month', nextPayment: 'Nov 1, 2025', status: 'Active' },
46   { service: 'Health Insurance', allocated: true, amount: '₹6,000 coverage', nextPayment: 'Mar 2026', status: 'Active' },
47   { service: 'Agnis Subsidy', allocated: false, amount: 'Pending', nextPayment: 'Verification Required', status: 'Pending' },
48   { service: 'Medical Aid', allocated: true, amount: '₹5,000 available', nextPayment: 'As needed', status: 'Active' },
49 ];
50
51 // Next actions for the user
52 const nextActions = [
53   { action: 'Visit Local Block Office', deadline: 'Within 7 days', priority: 'High', contact: '+91-1800-XXX-XXXX' },
54   { action: 'Submit Medical Certificate', deadline: 'Within 14 days', priority: 'Medium', contact: 'blockoffice@gov.in' },
55   { action: 'Update Bank Details', deadline: 'Within 30 days', priority: 'Low', contact: 'your bank branch id' },
56 ];
57
58 // Local support contacts
59 const supportContacts = [
60   { regionalOfficer: { name: 'District Social Welfare Officer', phone: '+91-944-2850-XXXX', email: 'dsw@gov.in', address: '123 Gandhi Road, Chennai - 600001' },
61     medical: { name: 'Government Hospital', phone: '+91-844-2852-XXXX', email: 'hospital@gov.in', address: '456 Anna Nagar, Chennai - 600046' },
62   };
63
64 const handleInput = (e) => {
65   e.preventDefault();
66   if (adhaarId.length !== 12) {
67     alert('Please enter a valid 12-digit Aadhaar Identification Number.');

```


[illegible]

```

1  =fÜÇ Starting Verification Test...
2
3  =föæ Logging in as Admin...
4  Γèà Login successful. Token received.
5
6  ΓÀ| Triggering Data Load (First Call - Expect delay)...
7  ΓÀñÀ First Call Duration: 3.88s
8  Γèà Data loaded. Records in response: 0 (Should be 5000)
9
10 ΓÜi Testing Cache (Second Call - Expect instant)...
11 ΓÀñÀ Second Call Duration: 0.01s
12 Γèà CACHE IS WORKING! (Response was instant)
13
14 =föÑ Testing Full Download Stream...
15
16 Γèà Download Complete!
17   - Total Bytes: 0.00 MB
18   - Time Taken: 0.03s
19   - Speed: 0.00 MB/s
20 ΓÜánÀ File size seems too small. Check if full data is being sent.
21

```

```

1      <!-- Unrelated Support (can be strong) -->
2      <!-- class="text text-gray-600">Personal assistance throughout the process</div>
3
4      </div>
5
6      </div>
7
8      <!-- class="flex items-start">
9      <div>
10         <span class="bg-green-600 text-white rounded-full w-6 h-6 flex items-center justify-center text-sm font-bold mb-3">•</span>
11         <div>
12             <strong>No More Service Denials</strong>
13             <p class="text-sm text-gray-600">Permanent solution for age-related biometric issues</p>
14         </div>
15     </div>
16
17     </div>
18
19     <!-- class="flex items-start">
20     <div>
21         <span class="bg-green-600 text-white rounded-full w-6 h-6 flex items-center justify-center text-sm font-bold mb-3">•</span>
22         <div>
23             <strong>Digital Inclusion Guarantee</strong>
24             <p class="text-sm text-gray-600">Ensures technology works for everyone, regardless of age</p>
25         </div>
26     </div>
27
28 </div>
29
30 </div>
31
32 </div>
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34 </section>
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[illegible]

B: Tables and Outputs

(Insert extra tables, charts, and intermediate results here)

NATIONAL ADMINISTRATIVE SUPPORT CELL

GOVERNMENT OF INDIA PORTAL

Technology Should Not Exclude You. Empowering 140 million senior citizens to overcome biometric failure through official diagnostic tools and administrative support.

TRACK APPLICATION STATUS

PORTAL FEEDBACK

PROTOCOL-2025

Authentication mismatch is a technical error, not a denial of rights.

XXXX XXXX 6208	VIHAAN SHETTY Age: 87	Chennai South, Besant Nagar	ADDRESS UPDATE	FINGERPRINT FAIL	32 mins ago	GENERATE DLC
XXXX XXXX 4263	VIKRAM PATEL Age: 69	Chennai South, T. Nagar	ADDRESS UPDATE	VERIFICATION	20 mins ago	GENERATE DLC
XXXX XXXX 5597	PRIYA MENON Age: 87	Chennai South, Mylapore	MANUAL EXCEPTION	VERIFICATION	2 mins ago	GENERATE DLC
XXXX XXXX 2618	AARAV FERNANDES Age: 61	Chennai South, Kodambakkam	ADDRESS UPDATE	FINGERPRINT FAIL	49 mins ago	GENERATE DLC

 **ELDERSSEVA**

RESOURCE CENTRE

GOVERNANCE



Upload Your Data

Upload your biometric CSV file to instantly generate analysis reports, detect duplicates, and visualize offline AI predictions.

[Click to Upload CSV](#)
Supported columns: State, District, Pincode, Age, BioAge, Status, FailureReason

All processing (including TensorFlow AI) happens locally on your browser. No data is uploaded.

Regional Data (57 states)			
State	Total	Success	Failures
Haryana	1635396	1635396	0
Bihar	4897576	4897576	0
Jammu and Kashmir	791217	791217	0
Tamil Nadu	4697967	4697967	0
Maharashtra	9225347	9225347	0
Gujarat	3196480	3196480	0
Odisha	2402756	2402756	0
West Bengal	2524294	2524294	0
Kerala	1609689	1609689	0
Rajasthan	3994905	3994905	0
Punjab	1739650	1739650	0
Himachal Pradesh	296216	296216	0



HOME

ANALYSIS

TRACK STATUS

FIELD AGENT

LEGAL RIGHTS

FEEDBACK

Data Analysis Report

File: api_data_aadhar_enrolment_500000_1000000.csv

Upload New File

Total Input

500000

Aadhaar Duplicates

0

Profile Duplicates

472307

Cleaned Data

27693

Quality Score

99%

Overview

Geographic

Trends

AI Anomaly Clusters

Show Duplicates (472307)

State	Total	Success	Failures
100000	1	1	0
Andhra Pradesh	2728	2469	259
Arunachal Pradesh	65	59	6

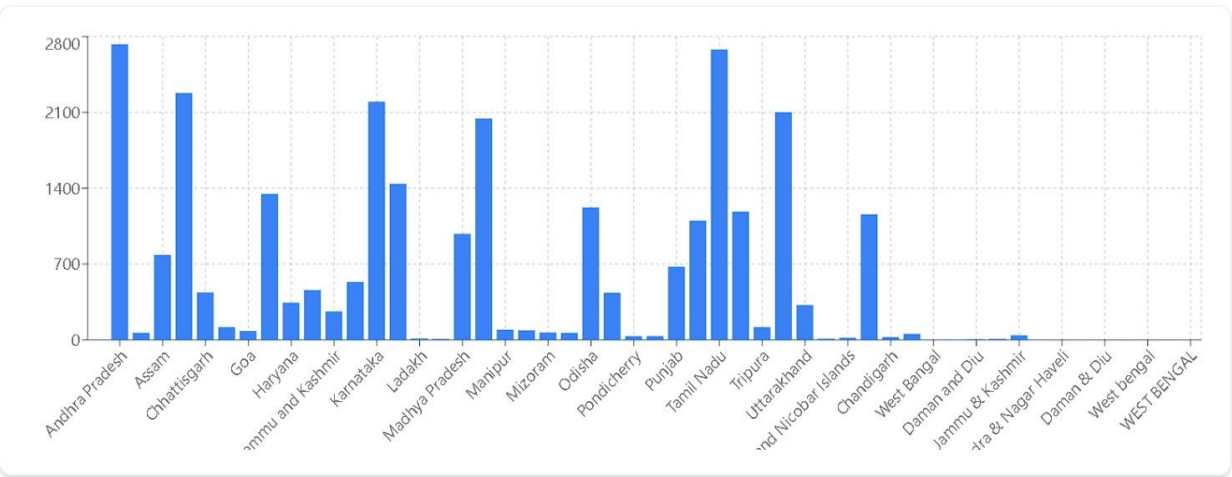
Overview

Geographic

Trends

AI Anomaly Clusters

Show Duplicates (472307)



State	Total	Success	Failures
100000	1	1	0
Andhra Pradesh	2728	2469	259

Westbengal	19	19	0
West Bengal	28	28	0
WEST BENGAL	37	37	0
West Bangal	54	54	0

← Previous

Page 1 of 2

Next →

 Admin Panel - Upload Custom Dataset

✓ Admin authenticated successfully!

Select CSV File to Upload

Choose file

No file chosen

Supported columns: state, district, pincode, age, bioAge (Smart detection enabled)

Logout

OFFICIAL VERIFICATION SYSTEM

IDENTITY STATUS TRACKER (DEMO)

Verify the real-time progress of your biometric exception or manual override status.

 **INDIVIDUAL IDENTITY SEARCH**
MANUAL OVERRIDE REGISTRY

● OFFLINE PROTECTION ENABLED

ID CITIZEN IDENTIFICATION NUMBER (AADHAAR)

REQUIREMENT: 12 NUMERICAL DIGITS ONLY

3 4 5 6 7 8 9 3 6 5 8 7

✓

ⓘ

PRIVACY PROTOCOL: AADHAAR NUMBERS ARE NEVER STORED ON PUBLIC SERVERS. SEARCH RESULTS ARE CACHED LOCALLY FOR OFFLINE REFERENCE.



ADMINISTRATIVE RECORD FOUND

Official record found for ID: *** 6587.**

Your request for 'Biometric Exception' is currently under 'Manual Field Verification'. A regional officer at your local block office is processing the override. Please keep your physical ID ready for any field visits.



BIOMETRIC TELEMETRY ANALYSIS

AUTOMATED SYSTEM DIAGNOSTICS



FINGERPRINT

FAILED

Elasticity Loss



IRIS SCAN

VERIFIED

High Precision



FACE MATCH

VERIFIED

Db Match Found

FINGERPRINT DECAY DETECTED

Primary authentication failed due to skin elasticity loss. However, secondary biometrics (Iris & Face) have confirmed identity.

RECOMMENDED SOLUTION

DOORSTEP "JEEVAN PRAMAAN" (DLC)

Via Postal 'Aadhaar Mitra' Agent

REF ID: DLC-REQ-2026-X89

Traceable via Elder Line 14567



CITIZEN: ZERO TRAVEL



GOV: GEO-TAGGED & SECURE

AUTO-ACTION: DISPATCHING DOORSTEP AGENT | REF: #DLC-REQ-2026-X89

● SYSTEM ACTION: REGULATION 14-B AUTO-TRIGGERED MANUAL OVERRIDE SERVICE

**RESOURCE ALLOCATION**
Benefits allocated to you

Old Age Pension

ACTIVE

₹2,500/month

Nov 1, 2025

Health Insurance

ACTIVE

₹30,000 coverage

Mar 2026

Ration Subsidy

PENDING

Pending

Verification Required

Medical Aid

ACTIVE

₹5,000 available

As needed

**NEXT STEPS**
After approval

ACTION REQUIRED	PRIORITY	DUE DATE	CONTACT
Visit Local Block Office	HIGH	WITHIN 7 DAYS	+91-1800
Submit Medical Certificate	MEDIUM	WITHIN 14 DAYS	blockoff
Update Bank Details	LOW	WITHIN 30 DAYS	Your bar ui



Application Submitted

OCT 12, 2025

Exception request registered in the National Portal.



Technical Audit

OCT 14, 2025

Automated check completed. Biometric mismatch confirmed due to age-related friction loss.

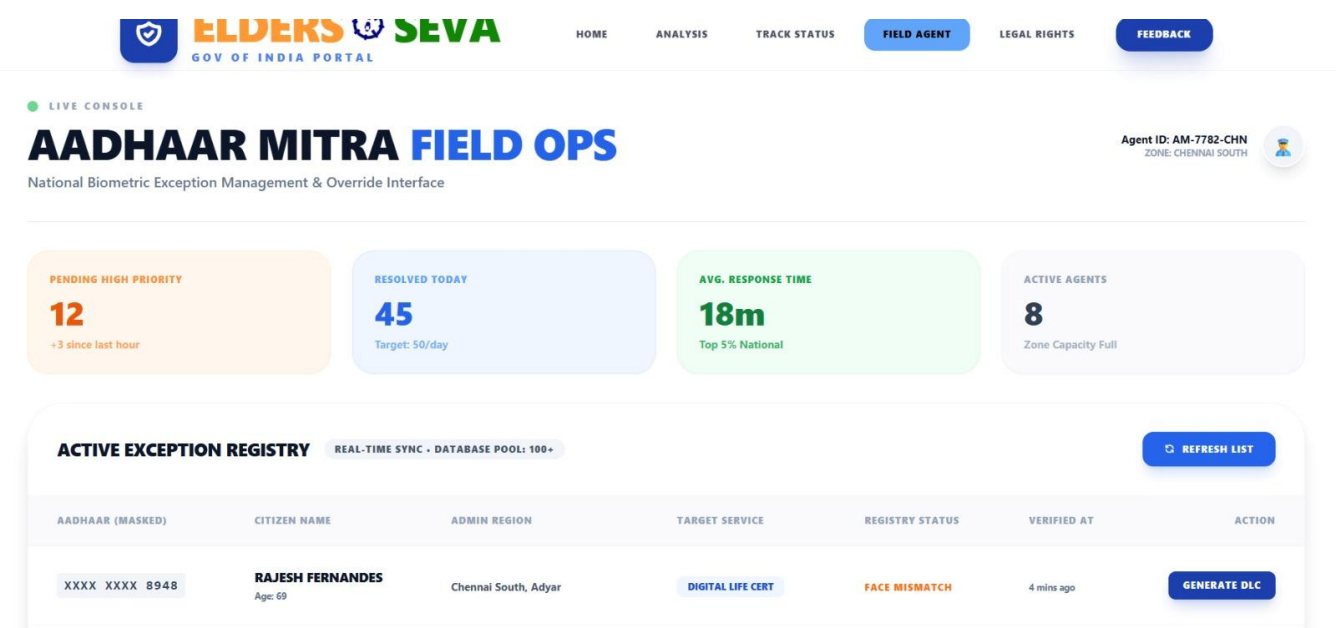


Manual Field Verification

OCT 16, 2025

Regional officer is currently verifying physical foundational documents.

ACTIVE MANUAL EXCEPTION ENABLED



Problem Solution;

EldersSeva addresses biometric obsolescence among senior citizens by introducing an intelligent biometric-exception framework that automatically detects age-related authentication failures and shifts the user to a secure manual verification flow. Once verified through geo-tagged doorstep visits or assisted identity checks, the citizen is permanently marked as biometric-exempt, ensuring uninterrupted access to pensions, healthcare, and welfare services without repeated biometric failures. The system supports offline continuity at service points and provides a centralized monitoring dashboard for authorities to ensure transparency, fraud prevention, and timely service delivery.

13. Conclusion :-

This project demonstrates how anonymized Aadhaar service usage data can be transformed into a powerful, privacy-preserving, policy-relevant decision-support system. By constructing a Migration Pressure Score, we show how administrative data can be used to anticipate service demand and support proactive governance.

Future work can extend this framework with additional datasets, predictive models, and deeper integration with administrative planning systems.